The Reward Is a Flock: A Two-Population Mean-Field Adversarial Game for Fleet Training

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Abstract

Reinforcement learning on language models is usually drawn as a policy optimizing against a reward. The reward is treated as furniture — a fixed function handed in from outside. This paper takes the reward as the object of study and argues it should be neither fixed nor singular. A single learned judge is a *proxy*: optimize it hard enough and Goodhart's law guarantees the policy finds where the proxy and the truth disagree. We replace the judge with a *flock* — a population of diverse judges whose aggregate verdict is the reward — and show three things. First, the flock is itself a mean-field system, but run in the *anti-ferromagnetic* regime: judges are rewarded for covering the judgment space, not for agreeing, so the flock's stable configuration maximizes independence, and independence is exactly what Condorcet's jury theorem needs and what shrinks the exploitable surface to the intersection of blind spots. Second, the generator fleet (the ferromagnetic policy population of the companion paper) and the discriminator flock couple through each other's mean fields with *opposite sign*: the pair is a two-population mean-field game, which is a generative adversarial network taken to the mean-field limit, and its equilibrium is a Nash *saddle* rather than a proxy peak. We make the anti-hacking claim formal: a static judge field admits the reward-hacking maximizer as a stationary point, and a co-evolving judge field does not, because the region the policy over-concentrates on is re-judged until the incentive to stay there vanishes. Third, none of this is safe without an *exogenous verifiable potential* — compile, test, benchmark, ship-and-stay-live — a common term on both populations that breaks the degenerate equilibrium in which flock and policy collude on a self-consistent but reality-detached reward. We show the whole construction collapses, in the discrete per-group limit, onto the group-relative advantage already computed by the shipped `hanzo-ml` GRPO trainer [5]: the flock's anchored verdict is the per-group mean field, and GRPO's existing weighted multi-reward sum is the non-adversarial, fixed-weight special case of a flock. We give an honest accounting of what is real in the training substrate today and what the adversarial, anchored, co-evolving generalization still needs.

1 The reward is the hard part

A policy-gradient trainer is only as good as the number it climbs. The companion papers treat the *population* of policies: how a fleet aligns (Paper 01, [1]) and how it searches (Paper 02, [2]). Both assume a field to align to and a fitness to select on. This paper is about that field. Where does it come from, and why is a single source of it dangerous?

The danger is Goodhart's law in its sharpest form. Let $q : \mathcal{X} \to \mathbb{R}$ be the true quality of an output x (unknown, possibly undefinable pointwise) and let $h : \mathcal{X} \to \mathbb{R}$ be a learned proxy for it. Optimizing $\max_x h(x)$ does not return the x that maximizes q; it returns the x that maximizes $h - q$ plus q, i.e. it seeks out and concentrates on the *disagreement* between proxy and truth. A capable policy is a very good optimizer, so it is a very good finder of that disagreement. This is reward hacking, and it is not a bug in a particular reward model — it is the generic behavior of maximizing a proxy.

Two structural moves defuse it, and this paper is about combining them. **Make the reward a population** (a flock of judges, Section 2), so that the exploitable disagreement must be shared by many independent proxies at once. **Make the reward move** (co-evolve the flock against the policy, Section 4), so that no static exploit is stationary. And underneath both, **anchor the reward to something real** (a verifiable potential, Section 6), so the flock cannot drift as a bloc.

2 The flock, and why independence is the whole game

Definition 1 (Flock reward)**.** Let $\mathcal{J}$ be a space of judges; a judge $j \in \mathcal{J}$ maps an output to a scalar verdict $j(x) \in \mathbb{R}$. Let $m_J \in \mathcal{P}(\mathcal{J})$ be a probability measure over judges (the *flock*). The flock reward is a functional of the flock,

$$r(x; m_J) \;=\; \mathrm{Agg}_{j\sim m_J}\, j(x),$$

with Agg the mean $\int j(x)\, dm_J(j)$ in the simplest case, and a robust statistic (median, trimmed mean, reliability-weighted mean) in practice.

The mean case exposes the essential fact. Write judge j's verdict as truth plus error, $j(x) = q(x) + \varepsilon_j(x)$, with $\mathbb{E}[\varepsilon_j] = b$ (a shared bias) and $\mathrm{Var}(\varepsilon_j) = \sigma^2$, and let $\bar\rho$ be the mean pairwise correlation of the ε_j across a flock of K judges.

Proposition 1 (Diversity is the reward's temperature)**.** *The variance of the mean flock reward is*

$$\mathrm{Var}\Big(\tfrac{1}{K}\textstyle\sum_{j=1}^{K} j(x)\Big) \;=\; \sigma^2\Big(\bar\rho + \tfrac{1-\bar\rho}{K}\Big) \;\xrightarrow[K\to\infty]{}\; \bar\rho\,\sigma^2.$$

Independent judges ($\bar\rho \to 0$) drive the variance to 0; perfectly correlated judges ($\bar\rho \to 1$) leave it at σ^2 — no gain from any number of them. The shared bias b is not *reduced by averaging at all; only decorrelated error is.*

Proof. The variance of an average of K unit-variance, mean-$\bar\rho$-correlated variables is $\frac{1}{K^2}\big(K + K(K-1)\bar\rho\big) = \bar\rho + \frac{1-\bar\rho}{K}$; scale by σ^2. The bias term is common to every judge and survives the average unchanged. □

Proposition 1 is the discriminative form of Condorcet's jury theorem [7]: a jury of weak but *independent* voters converges to certainty, and a jury of clones does nothing. Two consequences set the entire design.

Corollary 1 (The exploitable surface is an intersection)**.** *A policy hacks a single judge j by concentrating on x where $\varepsilon_j(x) > 0$. It hacks the flock only on x where $\varepsilon_j(x) > 0$ for enough j simultaneously — a set that shrinks with judge diversity, and vanishes for the components of error that are independent across the flock. Only the shared bias b (the common blind spot) remains hackable; Section 6 is how we remove it.*

Remark 1 (Judge diversity is Paper 01's temperature). Independence is not free: it must be produced. Judges drawn from one model family, one rubric, one persona, one decoding temperature are near-clones ($\bar\rho \to 1$). Diversity across model families, prompts, rubrics, and sampling temperatures is what keeps $\bar\rho$ low — and $\bar\rho$ plays exactly the role of the alignment order parameter m in Paper 01 [1]. A flock that has aligned into consensus is a flock frozen below its Curie point: coherent, and useless.

3 The flock is an anti-ferromagnet

If Paper 01 asks the policy population to align — a ferromagnetic mean field with coupling $J > 0$, whose stable state is consensus — then the flock must want the opposite. We give the judges a mean-field interaction with *negative* coupling.

Definition 2 (Anti-ferromagnetic flock objective)**.** Each judge minimizes a cost with three terms,

$$c_J(j; m_J, V) \;=\; \underbrace{\alpha\, \ell_{\text{ver}}\big(j, V\big)}_{\text{agree on the verifiable}} \;-\; \underbrace{|J| \int K(j, j')\, dm_J(j')}_{\text{repel on the subjective, } J<0} \;+\; \underbrace{\lambda\, \Omega(j)}_{\text{regularizer}},$$

where ℓ_{ver} penalizes disagreement with the verifiable potential V of Section 6, $K(j, j')$ is a similarity kernel on judges, and the middle term *rewards* being far from the rest of the flock (a repulsive, $J < 0$, mean-field interaction).

The flock's equilibrium is therefore *structured*: ferromagnetic (agreeing, $J > 0$ effectively) on the axis V can measure, and anti-ferromagnetic (diverse, $J < 0$) on the subjective axis it cannot. This is the correct anisotropy. On "does it compile," we want the judges to agree, because there is a truth to agree on. On "is the abstraction clean," we want them to spread, because the aggregate of diverse subjective reads is a better estimator than any one, and because a spread flock is unhackable where a consensus flock is a single point of failure. The repulsive term is what actively *produces* the low $\bar{\rho}$ that Proposition 1 needs; diversity stops being a hope and becomes an optimized quantity.

4 Two populations, opposite signs: the mean-field GAN

Now couple them. Let $m_\pi \in \mathcal{P}(\mathcal{X})$ be the policy population's output measure (the generator) and $m_J \in \mathcal{P}(\mathcal{J})$ the flock (the discriminator). Each population runs a mean-field game — a Hamilton–Jacobi–Bellman equation for the individual's optimal behavior against the population, coupled to a Fokker–Planck–Kolmogorov equation for how the population evolves under that behavior [6] — and the two are coupled through each other's mean fields.

The generator's individual cost *falls* with reward: $c_\pi(x; m_J) = -\, r(x; m_J) + \frac{1}{\beta}\, \mathrm{KL}(\pi \,\|\, \pi_{\text{ref}})$, where the KL-to-reference term is the temperature $1/\beta$ (Paper 01). The flock's cost *rises* with the reward's error against truth: it is paid to make $r(\cdot; m_J)$ track q (anchored by V) and to deny the generator any free lunch. The two objectives meet with opposite sign in r: the generator pushes m_π toward high r; the flock moves m_J to keep r honest exactly where m_π has gone.

Definition 3 (Two-population adversarial MFG)**.** An equilibrium is a pair (m_π^*, m_J^*) such that m_π^* is a mean-field best response to m_J^* in the generator game and m_J^* is a mean-field best response to m_π^* in the flock game. Equivalently it is a saddle of the min–max functional

$$\min_{m_J}\; \max_{m_\pi}\; \mathbb{E}_{x \sim m_\pi}\big[\, r(x; m_J) \,\big] \;-\; \gamma\, D\big(r(\cdot; m_J),\, q\big),$$

the generator maximizing the reward it is scored by, the flock minimizing the divergence D of that reward from truth.

This is a generative adversarial network [8] written at the mean-field limit: generator population vs. discriminator population, each a distribution rather than a single network. Its value is a saddle, not a maximum — and that distinction is the entire anti-hacking argument.

Theorem 1 (Existence under monotone coupling)**.** *Suppose the running costs are convex and Lipschitz in the control, the coupling functionals $m_\pi \mapsto c_J$ and $m_J \mapsto c_\pi$ are continuous for the Wasserstein topology and satisfy the Lasry–Lions monotonicity condition, and the state spaces are compact. Then the two-population system admits at least one equilibrium (m_π^*, m_J^*).*

Proof sketch. Each single-population game is a standard MFG whose best-response map is well defined and continuous under the stated assumptions [6]. The joint best-response $(m_\pi, m_J) \mapsto (\mathrm{BR}_\pi(m_J), \mathrm{BR}_J(m_\pi))$ is then a continuous self-map of the product of two convex, compact Wasserstein-ball measure sets; Schauder's fixed-point theorem gives a fixed point, which is by definition an equilibrium. Monotonicity gives uniqueness in the standard one-population regularity class; we do *not* claim uniqueness here, because the adversarial (opposite-sign) coupling is generically *non*-monotone — see Remark 2. □

Remark 2 (Honest boundary: non-monotone coupling can cycle)*.* The opposite-sign coupling that makes this a game rather than a potential-minimization is exactly what breaks Lasry–Lions monotonicity. Without the anchor of Section 6, the flow $(m_{\pi,t}, m_{J,t})$ can limit-cycle — the generator chases the reward, the flock chases the generator, neither settles — the mean-field echo of GAN training instability. The verifiable potential is what turns the cycle into a pinned saddle; it is not decoration, it is the well-posedness.

5 Why a moving reward cannot be hacked

Here is the payoff, stated as sharply as it deserves.

Proposition 2 (Static field admits the hack; co-evolving field does not)**.** *Fix the flock at m_J^0, so the reward is a static field $h(x) = r(x; m_J^0)$. The generator's problem $\max_{m_\pi} \mathbb{E}_{m_\pi}[h]$ is a linear program on $\mathcal{P}(\mathcal{X})$; its optimum places all mass on $\arg\max_x h(x)$. If $x^\star \in \arg\max h$ has $q(x^\star) < h(x^\star)$ — a proxy–truth gap, which by Corollary 1 exists wherever the flock's shared bias is nonzero — then the reward-hacking concentration on $x^\star$ is the* stationary point *of the generator dynamics.*

Now let the flock co-evolve: $m_J = m_J[m_\pi]$ is the flock's best response to the current policy population. The generator faces $\max_{m_\pi} \mathbb{E}_{m_\pi}\left[r(x; m_J[m_\pi])\right]$, in which the field is a functional of the very measure being optimized. The former stationary point $x^\star$ is not *stationary: as m_π concentrates on $x^\star$, the flock's best response (i) increases judge coverage near $x^\star$ (the anti-ferromagnetic repulsion drives judges into the newly dense, newly important region) and (ii) is pulled by the verifiable term toward $V(x^\star)$, which is low. Both lower $r(x^\star; m_J[m_\pi])$ until the generator's incentive to remain at $x^\star$ vanishes. No pure exploit is a fixed point of the coupled system; the only fixed points are saddles where the generator can no longer improve the reward without the flock immediately correcting it.*

The one-line reading: *a target that re-aims at wherever you aimed last cannot be shot with a fixed aim.* Reward hacking is optimization against a stationary proxy; remove the stationarity and you remove the hack. This is the formal content of "rotate the judges," "adversarial judges," and "the discriminator must keep up with the generator," unified as one property of the coupled equilibrium.

6 The anchor: a verifiable potential on both populations

A flock of *AI* judges has a failure mode that no amount of diversity fixes: it can drift as a bloc. If the shared bias b of Proposition 1 is itself learned and shared — a common sycophancy, a common taste, a hallucinated notion of quality — then the flock and the policy can settle into a self-consistent equilibrium that is internally coherent and externally false. In game terms, the min–max of Section 4 without an external term has a degenerate family of equilibria: any mutually-consistent (m_π, m_J) pair, including collapsed and reality-detached ones.

Definition 4 (Verifiable potential)**.** Let $V : \mathcal{X} \to \mathbb{R}$ be an *exogenous*, non-learned reward computable by execution: the code compiles; the tests pass; the benchmark clears its bit-exact gate; the change ships and stays live. V is a common potential added to *both* populations — it is a term in the flock's ℓ_{ver} (judges are penalized for verdicts inconsistent with V) and a term in the generator's reward.

Proposition 3 (Anchoring breaks the degeneracy)**.** *With V present, every equilibrium (m_π^*, m_J^*) must be consistent with V on the verifiable axis: the flock's V-inconsistent verdicts carry strictly positive cost, so no equilibrium can place mass on outputs that fail V while claiming high reward. The set of equilibria is pinned to the V-consistent manifold, eliminating the collapsed/hallucinated family and turning Remark 2's cycle into a bounded saddle.*

This is the RLVR (reinforcement learning from verifiable rewards) anchor married to RLAIF (reinforcement learning from AI feedback [10]): the *verifiable* axis is judged by execution, ferromagnetically (there is a truth, the judges should agree with it and each other); the *subjective* axis is judged by the anti-ferromagnetic flock. V does not have to be dense or complete — it only has to be *real* and *present*, enough to make reality a term the equilibrium cannot ignore.

Remark 3 (Judges graduate through the gate)*.* The co-evolution is closed by promoting judges: a fine-tuned enso model may *join* the flock, but only after clearing the verifiable gate on a held-out battery, and it is admitted with a reliability weight proportional to its historical agreement with V. A judge that keeps losing to the anchor is down-weighted. This is how the flock improves without eating its own tail: new judges enter through reality, not through consensus.

7 Both populations live at criticality; the GA moves them

Paper 01 argues each population should sit just below its ordering transition — coherent enough to converge, hot enough to correct. That applies to both here: a generator fleet frozen into one output has no exploration, and a flock frozen into one opinion has no robustness. The control knobs are the same temperatures — decoding temperature and KL coefficient for the generator, judge diversity $(1 - \bar{\rho})$ and coverage weight for the flock — annealed over training, never quenched. Paper 02's genetic search [2] operates on *both* populations: it evolves the policy variants (crossover of partial solutions) and it evolves the flock (recombine rubrics, retire drifted judges, breed diverse new ones). The three papers are therefore one coupled master system: a **generator fleet** $\otimes$ **discriminator flock**, **searched** by a genetic algorithm, **anchored** by a verifiable potential, all closing on a single mean-field number.

8 The grounding: it collapses onto GRPO

The construction above is not a proposal to build a new trainer. It is the adversarial, anchored, co-evolving generalization of the loop the `hanzo-ml` stack already runs. The shipped GRPO trainer [3, 5] samples a *group* of G rollouts per prompt, scores each, and forms the group-relative advantage

$$A_i \;=\; \frac{r_i - \text{mean}_{k \le G}\, r_k}{\text{std}_{k \le G}\, r_k} \qquad (\texttt{math::batch\_advantages},\ \text{ddof} = 1),$$

then a length-normalized policy-gradient loss with an optional $k3$ KL penalty to a frozen reference. Read this against Sections 2–4:

- r_i **is the anchored flock verdict** on rollout i: the aggregate of the diverse judges plus the verifiable term V. GRPO's existing "weighted multi-reward sum" (`score()`, `trainer.rs:119`) is precisely a flock with *fixed weights and no adversary* — the non-adversarial special case. Turning those fixed weights into a co-evolving, diversity-optimizing, V-anchored flock is the whole delta.
- **The group mean is the mean field.** $\text{mean}_k\, r_k$ is the finite-sample estimator of $\mathbb{E}_{m_\pi}[r]$; subtracting it is the mean-field coupling of Paper 01 computed per group. The std normalization is the local temperature. GRPO is, structurally, a per-group discretization of the mean-field game.
- **The $k3$ KL is $1/\beta$.** The frozen-reference penalty is the generator temperature that Section 4 keeps at criticality.

Cost. A flock of K judges is $K\times$ the eval cost per rollout. Two mitigations, both structural. *Cascade*: score with one cheap judge first; escalate to the full flock only on high stakes or when the cheap judge's uncertainty (or disagreement with V) is high. *Uncertainty gating*: the flock spread is an epistemic-uncertainty estimate (Proposition 1); train hard where the flock agrees and V confirms, and treat high-spread rollouts as exploration signal rather than gradient.

Where it lives. The eval-to-finetune bridge is the `/v1/evals` product (Hanzo's LLM-as-a-judge / observability plane): the flock scores every output there, the verifiers anchor it, and the aggregate is emitted as the reward stream the GRPO loop consumes. The judge population, its diversity objective, and its recalibration schedule are configuration on that plane, not new infrastructure.

9 What is real, what needs building, what is out of scope

Real (in the substrate today). The GRPO five-phase loop — rollout, weighted multi-reward, group-relative advantage with ddof = 1, length-normalized policy-gradient loss, $k3$ KL, AdamW step — is ported and tested [3]. The verifiable signals (V) exist as CI / harness outcomes (compile, test, benchmark gates, ship-and-live telemetry). The `/v1/evals` plane hosts LLM-as-a-judge scoring. Multi-model consensus primitives exist in the tool layer.

Needs building. The judge population as a first-class object: a diversity (anti-ferromagnetic) objective that actively minimizes $\bar{\rho}$; the co-evolution loop (flock best-responds to the policy population, drifted judges down-weighted, graduated judges admitted through the V gate); the cascade / uncertainty-gated escalation; and the wiring that turns the flock's anchored verdict into the per-group reward stream on `/v1/evals`. Today's "weighted multi-reward" is the fixed-weight, non-adversarial stub of the flock.

Out of scope. A full Lasry–Lions well-posedness proof for this specific non-monotone coupling (we invoke existence under monotonicity, Theorem 1, and are explicit that the adversarial coupling is non-monotone and can cycle without the anchor, Remark 2); learned *optimal* judge weights and kernels; and any human-in-the-loop calibration schedule beyond the verifiable gate.

10 Conclusion

The reward is not furniture. A single judge is a stationary proxy and will be hacked; a flock of diverse judges shrinks the exploitable surface to the intersection of blind spots (Proposition 1, Corollary 1), a flock that repels into diversity (anti-ferromagnetic, Section 3) actively produces the independence that shrinkage needs, a flock that co-evolves with the policy removes the stationarity that hacking requires (Proposition 2), and a verifiable potential pins the whole adversarial equilibrium to reality so the flock cannot drift as a bloc (Proposition 3). Structurally it is a two-population mean-field game — a GAN at the mean-field limit — and it collapses, per rollout group, onto the group-relative advantage the `hanzo-ml` GRPO trainer already computes, with today's fixed-weight multi-reward as its non-adversarial special case. Together with the fleet's ferromagnetic alignment (Paper 01) and the genetic search over both populations (Paper 02), it completes one system: *compass, engine, thermostat* — generator fleet $\otimes$ discriminator flock, searched, anchored, and closing on a single mean field.

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